Woodward (Buz) Galbraith

PhD Student, Khoury College of Computer Sciences, Northeastern University

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RESEARCH SUMMARY

PhD Student leveraging machine learning, databases, and biology to develop methods that turn heterogeneous biomedical data into structured, connected, and statistically reliable knowledge. Experienced with building systems for large scale data extraction and knowledge assembly, knowledge graph representations and biomedical entity linking. My current focus is on developing a system that automatically assembles ontology annotations over large scale data portals.

EDUCATION

PhD in Computer Science *Sept 2025 – Present*

Northeastern University, Khoury College of Computer Sciences, Boston, MA— GPA: 3.9/4.0

- **Advisor**: Dr. Benjamin M. Gyori, Gyori Lab for Computational Biomedicine
- **Honors**: ISMB Travel Fellowship ($500, 2026), Distinguished Research Fellow - Khoury College of Computer Sciences ($61,500, 2025)

M.S. in Data Science, focus in Biomedical Informatics *Sept 2022 – May 2024*

New York University, New York, NY — GPA: 3.9/4.0

- **Thesis:** A Knowledge Graph-Driven Approach to Understanding Cancer Risks and Treatment Recommendations under the advisement of Dr. Kushal Dey from the Memorial Sloan Kettering Cancer Center.
- **Relevant coursework:** Single-Cell Analysis, Proteomics, ML for Health Care, Big Data, Causal Inference

B.A. in Data Science and Economic Theory, Minor in Mathematics *Sept 2018 – May 2022*

New York University, New York, NY — GPA: 3.8/4.0

- **Honors:** Capital One Undergraduate Research Program ($8,000, 2022); Cum Laude (2022)

PUBLICATIONS

1. **Galbraith W**, Mammadov M, Pathak S, Gyori B, Pandey P. Similarity-Based Load-Balancing for Distributed Genomic Indices. *VLDB 2026 Workshop: Biomedical Data Management Systems (BioDMS)*, 2026.
2. Soni H, Nixon G, **Galbraith W**, Gyori B. Weakly Supervised Representation Learning for Cross-Ontology Mapping. *Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, 2026.
3. Arasu VA, Gadgil T, Rothstein JH, Alexeeff SE, Achacoso NS, Bhattacharya A, Cord JB, Esserman LJ, **Galbraith W**, Gerstley LD, et al., Habel LA, Sieh W. Comparative 10-year performance of mammography artificial intelligence, polygenic, and clinical breast cancer risk models in the Kaiser Permanente Research Bank. *JNCI: Journal of the National Cancer Institute*, djag154, 2026.
4. Ma X, Chow F, **Galbraith W**, Sultanov D, Behar EK, Hochwagen A. Evidence for strong purifying selection of human 47S ribosomal RNA genes. *PNAS* 123(16):e2529741123, 2026.
5. Chen S, Gupta N, **Galbraith W**, Shah V, Cirrone J. Prediction of drug effectiveness in rheumatoid arthritis patients based on machine learning algorithms. *ICBBE 22, Pages 146 – 154,* 2022.

SELECTED PROJECTS

DGLink: Automated knowledge graph construction over biomedical data portals

- Developed automated framework for assembling knowledge graph representations of biomedical data portals providing semantic interoperability over large scale data lakes.

- Collaborated with the National Cancer Institute (NCI) Cancer Research Data Commons and Sage Bionetworks to apply the DGLink framework across numerous portals covering heterogeneous data modalities, enabling cross-portal data discovery over the NCI Genomic, Proteomic, and General Data Commons through a combined knowledge graph representation.
- Implemented modular framework for annotating arbitrary experimental file types, including schema-matching based tabular data harmonization.
- LLM agents given DGLink KGs achieve up to 4× lower query time and cost, and improve F1 from 0.50 to 0.90, versus the same agent with only portal API access.

Similarity-Based Load-Balancing for Distributed Genomic Indices

- Implemented general-purpose load balancing strategy leveraging sketching data structures to efficiently distribute genomic sequences over distributed indices.
- Conducted comprehensive system benchmarking against baseline load balancing methods varying both number of files to be distributed and worker number.
- Found our method reduces index color table size by up to 8% and stored K-mer count by up to 15% over baseline.

Risk-Controlled Prediction Sets for Entity Linking

- Developed model agnostic framework leveraging conformal inference to reduce candidate set size while linking free text entity mentions to identifiers in standard resources while maintaining statistical risk control.
- Method achieves 50% candidate size reduction, while only reducing hits@5 from 0.75 to 0.73 for KRISSBERT model on BC5CDR benchmark dataset.

Weakly Supervised Representation Learning for Cross-Ontology Mapping

- Co-mentoring a group of NYU M.S. in Data Science students for their program capstone project.
- Developed weakly supervised RAG approach for identifying equivalent classes across ontologies leveraging SapBERT embedding model fine-tuned on SeMRA ontology mapping landscapes.
- Method improves hits@1 (0.79 to 0.93) over TF-IDF baseline on MONDO, MeSH ontology matching benchmark

RESEARCH EXPERIENCE

PhD Student, Gyori Lab for Computational Biomedicine *Sept 2025 – Present*

Northeastern University, Khoury College of Computer Sciences, Boston, MA

- Lead developer on DGLink, collaborating with NCI Cancer Research Data Commons and Sage Bionetworks on portal integration (see Selected Projects for technical details and results)
- Coordinate with lab collaborators on multiple side projects working on database systems, computational genomics and conformal prediction.
- Mentored numerous graduate and undergraduate researchers in the lab.

Machine Learning Engineer, Advanced Computational Infrastructure Team *May 2024 – May 2025*

Division of Research, Kaiser Permanente, Pleasanton, CA

- Consulted as a machine learning subject-matter expert across 10+ projects under six Principal Investigators; mentored undergraduate and graduate summer interns
- Optimized and containerized DICOM-based breast cancer risk models (Mirai) for the Radiomic and Genomic Predictors of Breast Cancer Risk project (PI: Dr. Laurel A. Habel); used GitLab CI/CD to ensure continuous model availability across 250,000+ patient records; contributed data curation and formal analysis to a resulting comparative study of AI, polygenic, and clinical risk models published in JNCI (2026)
- Led development of NLP models (traditional ML and LLM-based) for identifying breast cancer in clinical pathology reports under Dr. Lawrence Gerstley; best model achieved 0.952 AUPRC across 150,000 reports and was deployed as a RESTful API
- Administered a healthcare system-wide, PHI-heavy Kubernetes cluster supporting clinical and research workflows

Research Associate, Dey Lab *Sept 2023 – June 2024*

Computational & Systems Biology, Memorial Sloan Kettering Cancer Center, New York, NY

- Researched graphical machine learning methods (RotatE, TXGNN, GraphSAGE) for embedding and link prediction on a cancer-focused knowledge graph; project was a finalist for Meta's Lima Impact Grant
- Constructed multi-modal (demographic, genomic, mutation, treatment) knowledge graphs spanning 75,000+ cancer patients
- Developed inference models for cancer-relevant prediction tasks (e.g., cancer type from demographics, best treatment given cancer type), achieving a peak mean residual rank of 0.84/1.0 across tasks

Computational Researcher, Hochwagen Lab *Apr 2023 – July 2024*
Department of Biology, New York University, New York, NY
- Built a pipeline to download, filter, and process rRNA sequences from large public databases, constructing a dataset of 3,202+ humans' rRNA across diverse populations
- Analyzed rRNA sequence variance to identify highly conserved regions where mutations are likely to affect health; verified findings against simulated data with known mutations
- Contributed to a study of purifying selection on human 47S rRNA genes, published in PNAS (2026)

Summer Research Intern, Cirrone Lab *May – Sept 2022*
Department of Precision Medicine, NYU Langone Health, New York, NY
- Built a prediction model for individual patient response to biologic DMARDs using longitudinal data from 600+ rheumatoid arthritis patients
- Developed a novel two-stage ensemble model (severity regression followed by response classification), improving classification accuracy from 0.67 to 0.75 over baseline while increasing clinical interpretability

TALKS

- **Galbraith W**, Gyori B. DGLink: automated knowledge graph construction from biomedical data repositories. ISMB 2026 / BOKR, 2026

POSTER PRESENTATIONS

- **Galbraith W**, Li X, Gyori B, Han L. Risk-Controlled Prediction Sets for Entity Linking. *STAI-X 2026*, 2026
- **Galbraith W**, Gyori B. DGLink: automated knowledge graph construction from biomedical data repositories. SBHD 2026, 2026
- **Galbraith W**, Gyori B. DGLink: automated knowledge graph construction from biomedical data repositories. SWAT4HCLS 2026, 2026

SERVICE & TEACHING

- President, PhD Social Committee, Khoury College of Computer Sciences, Northeastern University, 2025–Present
- Capstone Mentor, *NYU Center for Data Science / Northeastern University*, 2025–Present
- Summer Intern Mentor, Kaiser Permanente Division of Research, 2024–2025
- Secretary, Out in STEM, NYU, 2022–2024
- Teaching Assistant, Survey Design, Analysis and Reporting, NYU School of Global Public Health, 2023
- Course Grader, Responsible Data Science, *NYU Center for Data Science*, 2022
- Head Course Grader, Mathematics for Economics I, *NYU Courant Institute of Mathematical Sciences*, 2022
- Course Grader, Mathematics for Economics II, *NYU Courant Institute of Mathematical Sciences*, 2021
- Secretary, Math Society, *NYU Courant Institute of Mathematical Sciences*, 2021–2022

TECHNICAL SKILLS

- **Programming:** Python, Rust, SQL, R, Bash, LaTeX, Java, PyTorch, TensorFlow, JAX, ONNX, Polars, Pandas
- **Databases:** Neo4j, MySQL, PostgreSQL, Redis, Oracle DB, Azure SQL, MongoDB, Elasticsearch
- **Infrastructure:** Git, Docker / Docker Compose, Kubernetes, Spark, Airflow, Ansible, GitLab CI/CD, SLURM / HPC cluster computing